

Targeted therapy

Targeted therapies are drugs that find and attack cancer cells. There are many different types of targeted therapy.

On this page

- [What is targeted therapy?](#)
- [Angiogenesis inhibitors](#)
- [Cancer growth inhibitors](#)
- [Monoclonal antibodies](#)
- [PARP inhibitors](#)
- [Testing suitability for targeted therapies](#)
- [Side effects of targeted therapies](#)
- [How we can help](#)

What is targeted therapy?

Targeted therapies are drugs that find and attack cancer cells. There are many different types of targeted therapy. Each one targets something in or around the cancer cell that is helping it grow and survive.

You can read more about the most common types below.

If you know the name of the drug you are looking for, you can use our [list of targeted and immunotherapy drugs](#) to find it. This gives more information about:

- what each treatment is
- how you have it
- possible side effects.



Angiogenesis inhibitors

Tumours need a blood supply to survive. Without a good blood supply, the tumour does not get the oxygen and nutrients it needs. This may slow the tumour's growth, or sometimes cause it to shrink.

Angiogenesis inhibitors block the chemical signals that cells use to make blood vessels grow. This makes it difficult for a tumour to develop the network of blood vessels it needs to get a blood supply.

Cancer growth inhibitors

Chemical signals tell cells in the body how and when to develop and divide. Cancer growth inhibitors make it difficult for cancer cells to receive these signals. By blocking the signals, these drugs may stop cancer cells from developing or dividing. This may slow the growth of a cancer.

Monoclonal antibodies

All cells in the body have protein receptors (targets) on their surface. A receptor is a bit like a lock. Another cell or substance can only connect to the receptor if it has the right key.

Monoclonal antibodies can only connect to one type of receptor. This means that they only target a certain type of cell, such as a cancer cell.

By connecting to the cancer cell's receptor, a monoclonal antibody could:

- block signals that tell cancer cells to grow and divide (also called a cancer growth inhibitor)
- block signals that help cancer cells develop a blood supply (also called an angiogenesis inhibitor)
- block signals that stop white blood cells attacking cancer cells (called a checkpoint inhibitor, and also described as a type of immunotherapy)
- help the immune system recognise cancer cells (also described as a type of [immunotherapy](#))
- block signals to protect the bones from damage caused by some types of cancer or cancer treatment – for example, [denosumab](#)
- carry a chemotherapy drug straight to the cancer cell.

A monoclonal antibody may also carry radiation directly to the cancer cell. This treatment is not widely used and may only be available as part of a [research trial \(clinical trial\)](#).

PARP inhibitors

PARPs are proteins that help damaged cells repair themselves.

PARP inhibitors block (inhibit) how PARP proteins work. This means cancer cells may become more damaged and die.

Some cancer cells have a genetic change called homologous recombination deficiency (HRD). PARP inhibitor drugs are even more effective against cells with HRD.

Testing suitability for targeted therapies

Before you have targeted therapy, you might have tests to find out if the treatment is likely to work for you.

The tests may check whether the cancer cells have a certain type of receptor for a monoclonal antibody to target.

You may have tests to look for genetic changes in your cancer cells. These include testing for HRD before PARP inhibitor treatment.

Doctors may check the cancer cells using a blood test, or by taking a sample of tissue (a [biopsy](#)). If you have had an operation, they may already have a sample of your tissue they can test. This kind of testing is not needed for all targeted therapies. If you need these tests, your doctor or nurse can tell you more.

Side effects of targeted therapies

Targeted therapies work differently to other cancer treatments such as chemotherapy. Different chemotherapy drugs often have similar side effects because they work in the same way.

The side effects of targeted therapy can be more varied. This is because the different types of drugs target cancer cells in different ways.

Some targeted therapy drugs can have more than one action. For example, a cancer growth inhibitor may also be an angiogenesis inhibitor.

Your specialist doctor, nurse and pharmacist will talk to you about the side effects of treatment.

Date reviewed

Reviewed: 01 August 2022 | Next review: 01 August 2025

! This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community



Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help